Angle sum and difference identities

1

a $\sin\alpha\cos\beta+\cos\alpha\sin\beta$

b $\sin\alpha\cos\beta-\cos\alpha\sin\beta$

2 $\dfrac{\tan\theta-\tan x}{1+\tan\theta\tan x}$

3 $\dfrac{\sqrt{3}\cos\theta+\sin\theta}{2}$

4 $\sin 3x\cos y-\cos 3x\sin y$

5 $\cos 2x\cos 3y-\sin 2x\sin 3y$

6

a $2-\sqrt{3}$

b $\dfrac{\sqrt{3}-1}{2\sqrt{2}}$

c $\dfrac{1-\sqrt{3}}{2\sqrt{2}}$

d $\dfrac{\sqrt{6}+\sqrt{2}}{4}$

e $\dfrac{\sqrt{2}\left(\sqrt{3}-1\right)}{4}$

f $\sqrt{3}+2$

g $\dfrac{\sqrt{6}+\sqrt{2}}{4}$

h $\dfrac{1}{2}$

i $\sqrt{3}$

j $\dfrac{\sqrt{3}}{2}$

7

a $-\cos x$

b $\sin x$

c $\cos x$

d $-\tan x$

e $\dfrac{\sqrt{2}\left(\cos x+\sin x\right)}{2}$

f $\dfrac{\cos x-\sqrt{3}\sin x}{2}$

g $\dfrac{1+\tan x}{1-\tan x}$

h $\dfrac{\sqrt{2}\left(\cos x+\sin x\right)}{2}$

8

a $\dfrac{364}{627}$

b $-\dfrac{644}{333}$

9

a $\cos x$

b $-\sin x$

c $\cos x$

d $-\tan x$

10 $2\cos\left(B-\dfrac{\pi}{2}\right)$

11

a $\dfrac{1}{2}$

b 1

c $\dfrac{1}{\sqrt{3}}$

12

a $\frac{4}{5}$ b $\frac{8}{17}$ c $-\frac{13}{85}$ d $\frac{84}{85}$

e Quadrant 4 f $-\frac{13}{84}$

13

a $\frac{4}{5}$ b $-\frac{5}{13}$ c $-\frac{63}{65}$ d $\frac{33}{65}$

e $-\frac{63}{16}$ f $-\frac{33}{56}$ g Quadrant 4 h Quadrant 2

14

a

i $\frac{\pi}{4}+\frac{\pi}{3}$ ii $\frac{\sqrt{2}-\sqrt{6}}{4}$

b

i $\frac{2\pi}{3}+\frac{\pi}{4}$ ii $\frac{\sqrt{6}-\sqrt{2}}{4}$

c

i $\frac{\pi}{3}-\frac{\pi}{4}$ ii $2-\sqrt{3}$

d

i $\frac{\pi}{4}-\frac{\pi}{3}$ ii $\sqrt{3}-2$

15

a $\sin(A+2B)$ b $\cos(6\theta+x)$ c $\tan 3m$ d $-2\sin\theta$

16

a $2\sin(5x)\cos(x)$ b $2\sin 4x\sin x$

c $2\cos(x+y)\cos\left(\frac{x-y}{2}\right)$ d $2\sin 31^\circ\cos 13^\circ$

17 $-\frac{\sqrt{3}}{\sqrt{2}}$

18 $x=\sin\theta,\ \cos\theta$

19 $x=20^\circ, 40^\circ, 140^\circ, 160^\circ, 260^\circ, 280^\circ$

Proofs

20 No solution provided.

21

a No solution provided.

b No solution provided.

c No solution provided.

d No solution provided.

e No solution provided.

f No solution provided.

g No solution provided.

h No solution provided.

22

a No solution provided.

b No solution provided.

23 No solution provided.

Applications

24

a 17 cm

b $17 \cos(\pi x)$

c 2 seconds

25

a 2.3 m

b 2 m

c $-\frac{1}{4} \sin \frac{\pi}{6} x + \frac{\sqrt{3}}{4} \cos \frac{\pi}{6} x + \frac{5}{2}$